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Voltiva 1.0 (Smart Plug)

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Abstract

The integration of Internet of Things (IoT) technologies has revolutionized energy management, enabling real-time monitoring and control of electrical systems. This project proposes an IoT-based energy measurement system tailored for residential and small-scale commercial use. The system continuously monitors electrical parameters such as current and voltage using reliable sensors like the ACS712 current sensor and the ZMPT101B voltage sensor. These data are processed and transmitted via an ESP32 microcontroller, leveraging its built-in Wi-Fi capabilities for seamless communication with a Firebase cloud database. The real-time data is accessible through a user-friendly mobile and web application, allowing users to monitor energy consumption patterns, detect anomalies, and make informed decisions to enhance energy efficiency. The system incorporates alert mechanisms to notify users of power down conditions.

Compared to existing solutions, this project emphasizes affordability, simplicity, and scalability. It bridges the gap in current energy monitoring solutions by offering a cost-effective and user-friendly alternative suitable for non-industrial applications. The calibration and optimization processes ensure accuracy and reliability, while the modular design facilitates future enhancements. By addressing the challenges of inefficient energy monitoring and the high cost of existing systems, this project contributes to sustainable energy management practices and empowers users to take proactive control of their energy usage.

Keywords: IoT-based energy monitoring; real-time data; ESP32; energy efficiency; microcontroller

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Abbreviations

ACS - Allegro Current Sensor

AC - Alternating Current

DC - Direct Current

ESP - Espressif Systems Protocol

IoT - Internet of Things

SCR - Silicon Controlled Rectifier

UI - User Interface

Wi-Fi - Wireless Fidelity

Chapter 1: Introduction

Energy consumption has become a critical concern in modern society, with increasing demand and rising energy costs necessitating efficient energy management solutions. Traditional energy monitoring systems are often costly, complex, and inaccessible for small-scale users, leaving a significant gap in the market for affordable and user-friendly alternatives. The advent of Internet of Things (IoT) technologies offers a promising avenue to address this issue by enabling real-time monitoring and control of electrical systems.

1.1 Research Background

This project focuses on developing an IoT-based energy measurement system tailored for residential and small-scale commercial use. Leveraging sensors such as ACS712 and **ZMPT101B**, alongside the **ESP32** microcontroller, the system ensures precise and reliable data acquisition. Real-time insights are made accessible through a web application, empowering users to track energy consumption, detect anomalies, and optimize energy efficiency. This approach not only reduces operational costs but also contributes to sustainable energy practices.

1.2 Objectives

The primary objective of this project is to develop an IoT-based energy measurement system capable of real-time monitoring of electrical parameters such as current and voltage. The system aims to provide users with seamless remote access through a web application to view energy data. Additional objectives include delivering insights into energy consumption patterns to help users make informed decisions to improve energy efficiency. This project also aims to incorporate detection capabilities to promptly alert users about power down conditions, ensuring efficient energy management. These objectives align with those outlined in the project proposal, emphasizing affordability, user-friendliness, and scalability for residential and small-scale commercial use.

1.3 Problem Statement

The lack of affordable and user-friendly systems for real-time energy monitoring and management presents a significant challenge, especially for residential and small-scale commercial users. Existing solutions are often expensive, overly complex, or tailored for industrial-scale applications, leaving a gap in the market for simpler alternatives. Furthermore, many users struggle to identify inefficient energy usage patterns in their electrical systems, which can lead to higher energy bills and potential equipment failures. This project addresses these issues by proposing a cost-effective IoT-based solution that continuously monitors energy parameters and provides actionable insights to users. The proposed system will empower users with data to enhance energy efficiency while reducing costs and improving safety, consistent with the problem statement in the project proposal.

1.4 Scope and Limitations

The scope of this project includes developing an IoT-based energy monitoring system tailored for residential and small-scale commercial applications. It involves the use of ACS712 current sensors and ZMPT101B voltage sensors for accurate data collection, integrated with an ESP32 microcontroller for real-time data transmission. The system supports remote access via web application and provides basic insights into energy usage patterns. However, the project is not designed for large-scale industrial applications and does not include advanced features like predictive maintenance or machine learning-based analytics. The software is limited to real-time data visualization and historical data review without incorporating complex control functions. These limitations ensure the solution remains affordable and user-friendly, aligning with the scope and boundaries detailed in the project proposal.

Chapter 2: Literature Review

The advancement of IoT-based energy monitoring systems has gained significant attention in recent years due to the rising need for efficient energy management in residential and commercial applications. Traditional energy monitoring systems, such as analog and digital meters, are limited in functionality, offering only basic readings without real-time data accessibility or remote monitoring capabilities. These standalone devices are often inconvenient for users managing multiple locations and lack advanced analytics to provide actionable insights. Although suitable for basic applications, they fail to meet the growing demand for integrated and efficient energy management solutions.

In contrast, modern IoT-based systems integrate advanced sensors, microcontrollers, and cloud platforms, enabling real-time monitoring and control. For example, IoT systems utilizing Arduino and GSM modules allow energy data transmission to central servers, enhancing accessibility. Similarly, solutions based on Raspberry Pi provide accurate data visualization and advanced analytics. However, despite their effectiveness, these systems often come with high costs and complexity, making them less feasible for residential and small-scale commercial applications.

Current sensor technologies, such as the ACS712 current sensor and the ZMPT101B voltage sensor, are widely used for their affordability and accuracy. These sensors are well-suited for small-scale energy monitoring systems, but challenges like calibration difficulties and sensitivity to environmental conditions limit their performance. Communication protocols also play a critical role in IoT-based systems. MQTT, known for its lightweight and efficient real-time communication, is commonly used in energy monitoring applications but requires optimization to minimize latency and power consumption. HTTP, another popular protocol, offers simplicity and broad support but is less efficient for IoT applications due to its higher overhead.

Despite these advancements, significant research gaps persist. Existing solutions often prioritize industrial and large-scale applications, leaving a need for affordable and accessible systems tailored to smaller users. Scalability is another challenge, as most systems lack modular designs that allow for easy expansion without significant reconfiguration. User interfaces are also frequently underdeveloped, limiting accessibility for non-technical users. Furthermore, integrating advanced features such as AI-driven analytics for predictive maintenance and anomaly detection remains an underexplored area in small-scale applications. Additionally, improving the adaptability of sensors to harsh environmental conditions can enhance the reliability and accuracy of energy monitoring systems.

By addressing these research gaps, the proposed system aims to bridge the divide between affordability, scalability, and usability. It focuses on delivering a cost-effective, user-friendly solution equipped with real-time monitoring capabilities and advanced features. This system has the potential to set a new standard in IoT-based energy management for residential and small-scale commercial users, paving the way for further innovation and research in this domain

2.1 Existing Solutions

2.1.1 Traditional Energy Monitoring Systems

Traditional energy monitoring systems primarily rely on standalone devices such as analog or digital energy meters, which can provide basic readings like voltage, current, and power consumption. However, these systems have significant limitations:

- **Limited Data Access**: Traditional systems lack real-time data accessibility, requiring physical access to the device to retrieve readings.
- **No Remote Monitoring**: Users cannot monitor energy consumption remotely, which is particularly inconvenient for managing multiple properties or distributed systems.
- **Absence of Advanced Analytics**: These systems do not offer insights like usage trends, anomaly detection, or predictive analysis.
- **Cost and Scalability Issues**: While suitable for simple applications, traditional systems do not support easy integration with modern energy management practices or other IoT-based systems.

2.1.2 Advanced IoT-Based Solutions

Modern IoT-based energy management solutions address many limitations of traditional systems by integrating sensors, microcontrollers, and cloud platforms to enable real-time monitoring and control. Examples include:

- **IoT-Based System with Arduino and GSM Modules**: A system designed in [1] utilized Arduino microcontrollers paired with GSM modules to transmit energy data to a central server. While this approach enhanced data accessibility, it faced constraints in scalability and required GSM network availability.
- **Energy Monitoring Framework Using Raspberry Pi**: In [2], researchers proposed a solution based on Raspberry Pi, integrating high-resolution sensors and providing advanced data visualization. However, the system's high initial cost and complexity limited its suitability for small-scale users.

Despite these advancements, challenges remain in achieving an optimal balance between cost, scalability, and usability, particularly for small-scale residential and commercial application

2.2 Review on Existing Methodologies

2.2.1 Sensor Technologies

Current sensors such as ACS712 and voltage sensors like ZMPT101B are widely recognized for their accuracy, affordability, and ease of integration into IoT-based systems [3]. The ACS712 sensor measures current by detecting the magnetic field generated by current flow, while the ZMPT101B sensor captures voltage levels with high precision. Despite their advantages, these sensors have limitations, such as susceptibility to temperature variations and electromagnetic interference, which can impact their performance. Calibration processes are critical to ensure accuracy, but they can be time-consuming and require expertise, presenting a barrier to widespread adoption. Research highlights the need for advanced calibration techniques and sensor designs that mitigate environmental sensitivities and enhance durability

2.2.2 Communication Protocols

Efficient communication protocols are vital for seamless data transmission between IoT devices and cloud platforms. Two widely used protocols are:

- **MQTT (Message Queuing Telemetry Transport)**:
    - Features: Lightweight and ideal for constrained networks, MQTT enables reliable and efficient real-time communication.
    - Challenges: Integration with hardware systems, especially in resource-constrained environments, requires optimization to minimize latency and energy consumption [4].
    - Use Cases: Frequently employed in systems where bandwidth efficiency and low power usage are critical.
- **HTTP (Hypertext Transfer Protocol)**:
    - Features: A ubiquitous protocol offering simple implementation for sending and receiving data over the internet.
    - Challenges: While HTTP is widely supported, it is less efficient for IoT applications due to its higher overhead compared to MQTT.
    - Use Cases: Best suited for applications where data size and communication frequency are less critical.

Chapter 3: Theory and Methodology

3.1 Theory

Voltage and Current: Voltage is the potential difference across a circuit, while current represents the flow of electric charge.

Power: The product of voltage and current. Real power (in watts) is the actual power consumed by a device.

Energy Consumption: The total power used over time, measured in kilowatt-hours (kWh).

3.1.1 Voltage Calculation

The voltage sensor (e.g., ZMPT101B) outputs an analog signal proportional to the measured voltage. Use the following equation:

$$V = \left(\frac{ADC\ Value}{ADC\ Resolution}\right) \times V_{ref} \times K_{voltage}$$

- **ADC Value**: The raw analog-to-digital conversion value from the ESP32.
- **ADC Resolution**: The maximum ADC value (e.g., 4095 for 12-bit ADC).
- **V_{ref}:** The reference voltage of the ADC (e.g., 3.3V for ESP32).
- **$K_{voltage}$:** Calibration constant (voltage divider factor).

3.1.2 Current Calculation

The current sensor (e.g., ACS712) outputs an analog signal with a baseline voltage Representing zero current. Use the following equation:

$$I = \left(\frac{\text{Vout} - \text{Voffset}}{\text{Kcurrent}}\right)$$

- **V_{out}:** The voltage output from the sensor (calculated from ADC value).
- **V_{offset}:** The sensor's zero-current voltage (e.g., 2.5V for ACS712).
- **$K_{current}$:** Current sensor sensitivity (e.g., 0.185V/A for ACS712-5A).

3.1.3 Power Calculation

Real power (P) is the product of voltage and current:

$$P = V \times I \times Cos\emptyset$$

cosϕ: Power factor (1 for purely resistive loads; less for inductive or capacitive loads.

3.1.4 Energy Consumption Calculation

Energy (E) is the integral of power over time:

$$E = \int P\, dt$$

In a discrete system (like using Arduino), approximate this as:

$$E = \int \sum (P \times \Delta t)$$

Δt: Time interval between measurements (in hours for kWh).

3.2 Overview of the Methodology

This methodology outlines the design, implementation, and testing of a smart plug system for energy monitoring and power-down alerting. The system incorporates sensors, a relay-based backup power system, and a GSM module to notify users of power outages in real time.

1. System Design

The project integrates:

- **Sensing Unit**: Voltage and current sensors measure electrical parameters in real-time.
- **Control Unit**: An ESP32 microcontroller processes sensor data and manages communication and alerts.
- **Backup Power System**: A relay-based system ensures the GSM module remains powered during outages.
- **Communication Unit**: A GSM module sends SMS alerts to users when the main power fails.

- **Cloud Integration**: The ESP32 uploads energy data to a cloud platform for analysis.

2. Circuit Design and Hardware Setup

- **Voltage and Current Measurement**:
  - The ZMPT101B voltage sensor and ACS712 current sensor capture electrical parameters.
- **Backup Power System**:
  - A relay module switches between the main power supply and the backup battery.
  - The GSM module is powered by the backup battery when the relay detects a power failure.
- **GSM Module**:
  - A SIM800L module is used to send SMS alerts to the user. The ESP32 triggers the GSM module when power loss is detected.
- **ESP32 Microcontroller**:
  - Acts as the central unit, managing sensor data, backup power switching, and GSM alerts.

3. Arduino Program (For ESP32) Development

The software is developed in the Arduino IDE and performs the following tasks:

- **Data Acquisition**:
  - Captures voltage and current sensor readings and converts them to meaningful metrics.
- **Power-Down Detection**:
  - Monitors voltage and current levels to identify power failures.
  - Activates the relay to connect the backup battery to the GSM module.
- **Alerting**:

    - Sends an SMS notification to the user using the GSM module when a power outage occurs.
    - The alert includes the timestamp and system status.

4. Web Application Development

The web application serves as the primary interface for users to monitor and analyze energy data:

- **Front-End Development**:
    - Designed using **HTML** and **CSS** for a responsive and user-friendly interface.
    - Displays energy consumption data, voltage, current readings, and logs of power failures.
- **Back-End Development**:
    - Built using **PHP** to handle data processing and server-side logic.
    - Integrates with a **MySQL** database to store energy metrics and system events.
- **Database Design**:
    - Tables are created for logging sensor data, power failure events, and user information.
- **Visualization**:
    - Graphs and tables display energy trends and system status for easy interpretation.

3.3 Calibration

Calibration is an essential step in ensuring the accuracy and reliability of the measurements taken by sensors in an energy monitoring system. It ensures that the raw sensor data is converted into meaningful, real-world values such as voltage, current, and energy.

3.3.1 Voltage Sensor (ZMPT101B) Calibration

The **ZMPT101B** voltage sensor was used to measure AC voltage. Calibration was performed by comparing the sensor's output voltage with a known accurate reference voltage.

Calibration Steps:

1. **Set Up Equipment**:
   - A **calibrated multimeter** was used to measure the true AC voltage.
   - The ZMPT101B voltage sensor was connected to the system, and it was attached to the ESP32.
2. **Measure Sensor Output**:
   - The sensor's output voltage was read through the ESP32's analog input pin. The output signal was a proportional representation of the AC voltage being measured, but it had to be scaled to the actual AC voltage.
3. **Calculate the Conversion Factor**:
   - The ZMPT101B sensor provided an output voltage that was proportional to the input AC voltage. Typically, it was known that the sensor provided a 0–5V output for the range of 0 to 250V AC.
   - For example, if the sensor output was 2.5V for 120V AC, the conversion factor was calculated as:

$$\text{Conversion Factor} = \frac{120}{2.5} = 48$$

4. **Validate Calibration**:
    - The sensor readings were compared with the true voltage readings from the multimeter.

3.3.2 Current Sensor (ACS712) Calibration

The **ACS712** current sensor was used to measure the current flowing through the load. Calibration was necessary because the sensor's output had an offset when no current was flowing.

Calibration Steps:

1. **Zero Current Adjustment**:
    - The ACS712 sensor gave an output voltage centered around 2.5V when no current flowed through the sensor. Calibration was done to ensure that the output voltage was exactly 2.5V when no current was flowing. Any discrepancies were adjusted in the software by applying an offset.
2. **Apply Known Loads**:
    - A known resistive load, such as a 100W incandescent bulb, was connected to the circuit. The true current was measured using a **calibrated multimeter** in series with the load.
    - The output voltage from the ACS712 sensor was recorded.
3. **Calculate the Conversion Factor**:

    The ACS712 sensor provided an output voltage change of 185mV per ampere (for the 5A version).

    The formula to convert the sensor's output to current was

$$I = \frac{Vout - 2.5}{185\ mV/A}$$

For example, if the sensor output was 2.7V, the current was calculated

$$I = \frac{2.7-2.5}{0.185} = 1.08A$$

This conversion factor was adjusted in the code to ensure accurate Current readings.

4. **Validate Calibration**:

- The current reading from the ACS712 was compared with the true value from the multimeter.

3.4 Optimization

3.4.1 Sensor Data Optimization

Sampling Rate Adjustment:

- The sensor data sampling rate was optimized to balance between real-time accuracy and power consumption. By adjusting the sampling frequency, the system was made efficient, ensuring that data was captured at the right intervals without overwhelming the microcontroller.
- A lower sampling rate (e.g., once every second or every few seconds) was used for voltage and current sensors, as excessive sampling could lead to unnecessary power consumption and data overload.

3.4.2 Data Transmission and Communication Optimization

Efficient Data Uploading (ESP32 to Web Server):

- Data transmitted from the ESP32 to the web server was optimized by compressing the data to reduce the bandwidth required and improve transmission speed. For instance, instead of sending raw sensor readings, only the processed values (voltage, current, energy, and status) were sent.

- A **retry mechanism** was implemented in the code, which ensured that data was successfully transmitted to the web server in case of temporary network issues. This reduced the risk of data loss due to communication failures.

- In case of a power failure, data integrity was maintained by saving data locally on the ESP32's memory. Once power was restored, the saved data was transmitted to the web server to ensure no data loss.

3.4.3 Web Application Optimization

Database Optimization:

- The MySQL database used for storing sensor data was optimized by indexing frequently queried fields (e.g., date/time, power consumption) to speed up data retrieval and reduce query execution time.

- SQL queries were optimized by limiting the number of records retrieved, using **pagination** for displaying large sets of data, and avoiding complex joins. This improved the web application's responsiveness.

Front-End Optimization

- The web application was optimized for better user experience by implementing **lazy loading** for displaying graphs and tables. Instead of loading all data at once, only the required portion of data was loaded as needed, reducing page load times.

- The web application was designed to be fully responsive, ensuring that it could be accessed on various devices (PCs, tablets, and smartphones). The user interface was optimized to fit different screen sizes, improving the user experience.

Chapter 4: Results

4.1 Modular Test Results

4.1.1 Voltage Sensor Calibration Test Data

Test Voltage (V)	Measured Voltage (V) (Multimeter)	Measured Voltage (V) (ZMPT101B)	Discrepancy (%)
120	120.0	120.1	0.08
230	230.0	229.9	0.04
100	100.0	100.2	0.20
50	50.0	49.8	0.40
150	150.0	149.6	0.13
200	200.0	199.7	0.15
80	80.0	80.2	0.25
120	120.0	120.3	0.25
230	230.0	230.1	0.04
100	100.0	100.1	0.10

Table 4.1 Voltage Calibration Test Data

4.1.2 Current Sensor Calibration Data

Test Current (A)	Measured Current (A) (Ammeter)	Measured Current (A) (ACS712)	Discrepancy (%)
0.5	0.5	0.51	2.00
2.0	2.0	2.01	0.50
3.5	3.5	3.49	0.29
5.0	5.0	5.01	0.20
0.2	0.2	0.19	5.00
0.2	0.2	1.02	2.00
2.5	2.5	2.49	0.40
4.0	4.0	3.98	0.50
0.8	0.8	0.82	2.50
3.0	3.0	2.98	0.67

Table 4.2: Current Calibration Test Data

4.2 System Test Analysis

The **System Test Analysis** aims to evaluate the overall performance of the energy monitoring system, ensuring that all individual components and modules work together as expected to achieve the desired functionality.

- **Real-Time Data Collection**:
    - Voltage and current sensors were tested with a range of voltages and currents to ensure proper operation.
- **Power Failure Simulation**:
    - A power-down event was simulated by disconnecting the main power supply, and the system was monitored for a proper transition to backup battery power.
    - The relay-based backup power system was tested to ensure smooth switching between the mains and backup battery.
- **GSM Module Functionality**:
    - A GSM test was conducted by simulating a power failure and checking if an alert was sent to the user.
    - The GSM module was also tested for power consumption during power-down events.
- **Web Application Testing**:
    - The web application was checked for real-time data retrieval and display.
    - Historical data was queried to ensure the application provided accurate and timely information.
- **End-to-End System Performance**:
    - All components were integrated, and the system was run as a whole.

Chapter 5: Discussion and Conclusion

5.1 Discussion

The proposed project, a smart plug with real-time voltage, current, and power measurement, daily energy consumption reporting through a web app, and a power-down alert SMS feature, represents an innovative step toward enhancing energy management at the residential and small-scale commercial levels. The integration of IoT technology with energy monitoring solutions brings numerous advantages, offering users better control, insights, and efficiency in managing their electrical systems.

The system's capability to measure voltage, current, and power in real time addresses a crucial need for continuous monitoring of electrical parameters. By utilizing reliable sensors such as the ACS712 for current measurement and the ZMPT101B for voltage measurement, the system ensures accurate and safe monitoring. Real-time data not only enables users to identify inefficiencies promptly but also aids in diagnosing potential electrical faults, such as overcurrent conditions or voltage fluctuations. These features are essential for protecting electrical appliances and ensuring the safety of the electrical system.

Providing users with daily energy consumption reports via a web application adds significant value to the project. By leveraging cloud platforms like Firebase for data storage and synchronization, the system ensures real-time updates and seamless access to historical energy data. These reports empower users to analyze their energy usage patterns, identify areas of wastage, and make informed decisions to optimize energy efficiency. Such insights are particularly beneficial for small businesses and homeowners aiming to reduce energy costs and improve sustainability.

The web application developed using Node.js for the backend and React for the frontend ensures an intuitive and responsive user interface. These technologies enable real-time data visualization, making it easier for users to monitor and manage their energy usage remotely. The inclusion of basic user authentication further ensures that only authorized users can access the data, enhancing system security.

One of the standout features of this project is the power-down alert via SMS. This functionality ensures that users are immediately notified of power outages, allowing them to take timely action to prevent equipment damage or data loss. By integrating SMS gateways, the system bridges the gap between IoT monitoring and direct user communication, offering a practical solution for critical notifications.

The implementation of this project may face several technical challenges, including ensuring accurate sensor calibration, maintaining data integrity during transmission, and achieving low-latency alerts. To address these challenges, the project adopts well-established components such as the ESP32 microcontroller for its dual-core processing power, built-in Wi-Fi, and low energy consumption. The compatibility of ESP32 with the Arduino IDE simplifies programming and debugging, reducing development time.

While the project is designed for residential and small-scale commercial applications, its modular design allows for potential scalability. Additional features, such as integration with predictive maintenance algorithms or support for more complex IoT ecosystems, could be added in future iterations. However, the current focus on simplicity and cost-effectiveness ensures that the system remains accessible to a wide range of users without requiring extensive technical expertise.

The proposed smart plug system demonstrates a practical and user-friendly approach to real-time energy monitoring and management. By combining advanced IoT technologies with essential features like real-time data access, energy consumption reporting, and power-down alerts, the system addresses a critical gap in affordable and efficient energy management solutions. This project not only enhances the user's ability to oversee their electrical systems but also contributes to broader efforts toward energy conservation and sustainability.

5.2 Conclusion

The research project successfully developed a smart plug system capable of real-time monitoring of voltage, current, and power, providing users with daily energy consumption reports and proactive notifications for power outages. These features empower users to optimize their energy usage, improve efficiency, and enhance the safety of their electrical systems. The system's integration of IoT technology ensures cost-effectiveness and accessibility, making it a viable solution for residential and small-scale commercial applications.

5.3 Future Work

To improve the current solution, several enhancements can be considered. First, the integration of predictive maintenance algorithms could allow the system to anticipate potential faults before they occur. Second, incorporating machine learning techniques could provide advanced analytics, such as anomaly detection and energy usage forecasting. Additionally, expanding the system's compatibility with smart home ecosystems, such as Alexa or Google Home, could improve user convenience and adaptability. Lastly, developing a mobile application alongside the web interface would enhance accessibility and user experience, ensuring seamless interaction with the smart plug system.

Chapter 6: References

[1] X. Zhang and Y. Wang, "IoT-enabled real-time monitoring system for electrical consumption," *IEEE Internet of Things Journal*, vol. 6, no. 4, pp. 7451-7459, Aug. 2019.

[2] J. Brown et al., "A comparative analysis of microcontroller platforms for IoT systems," *IEEE Transactions on Consumer Electronics*, vol. 65, no. 2, pp. 127-135, May 2020.

[3] M. K. Lee, "Review of current sensors for smart metering applications," *IEEE Sensors Journal*, vol. 19, no. 6, pp. 2393-2403, Mar. 2019.

[4] L. Singh and P. Kumar, "Design and performance analysis of voltage sensors in IoT projects," *IEEE Access*, vol. 7, pp. 45100-45109, 2019.

[5] H. Alvarez et al., "Cloud integration for IoT data management using Firebase," *IEEE Cloud Computing*, vol. 7, no. 3, pp. 24-34, June 2020.

[6] S. R. Patel and A. Kumar, "Evaluating real-time databases for IoT applications," *IEEE Access*, vol. 8, pp. 90356-90364, 2020.

[7] Y. Zhao and L. Huang, "Scalable server architecture for real-time applications using Node.js," *IEEE Transactions on Services Computing*, vol. 13, no. 5, pp. 877-889, Sept. 2020.

[8] P. Smith and T. Davis, "React's efficiency in modern web applications," *IEEE Software*, vol. 38, no. 1, pp. 22-29, Jan. 2021.

[9] C. Wilson, "Development tools for IoT: A comparative study of Arduino IDE and its competitors," *IEEE Embedded Systems Letters*, vol. 12, no. 4, pp. 97-101, Dec. 2020.